DAY 19 | POST-READ

Deploying to Azure: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

Part A: Summary of today

Topic	What to remember
Cloud-ready app	Config from environment variables, stateless, /health and /ready, logs to stdout.
docker compose	Several services in one file; services reach each other by name; volumes persist data.
ACR	Azure Container Registry stores images; az acr build builds in the cloud.
App Service	PaaS for web apps; Linux plan; deploy code or a container; set WEBSITES_PORT.
Container Apps	Serverless containers; az containerapp up --source; scales automatically, even to zero.
AKS	Full Kubernetes; choose only with a clear need and a team to run it.
CD pipeline	Test, build and push, deploy dev, approval, deploy prod with the same image tag.
Environments	Hold approvals and deployment history in Azure DevOps.
Safe habits	Tag by build id, smoke test after deploy, secrets in Key Vault, clean up.

Part B: Going beyond today

1. Pulling from a private registry securely

Give the App Service or Container App a managed identity and grant it the AcrPull role on the registry. No registry passwords stored anywhere.

2. Secrets with Key Vault references

```
az webapp config appsettings set -g rg-demo -n tasks-api-demo123 \
  --settings "DATABASE_URL=@Microsoft.KeyVault(SecretUri=https://kv-demo.vault.azure.net/secr
```

The app still reads an environment variable; the value comes from Key Vault at runtime.

3. Deployment slots

App Service slots let you deploy to a staging slot, warm it up, and swap it with production almost instantly. Swapping back is your rollback.

4. Scale rules in Container Apps

```
az containerapp update -n tasks-api -g rg-demo --min-replicas 0 --max-replicas 5 \
  --scale-rule-name http --scale-rule-type http --scale-rule-http-concurrency 50
```

5. Smoke tests in the pipeline

```
- script: curl --fail --retry 5 --retry-delay 10 https://$(appUrl)/health
  displayName: Smoke test
```

If the new version does not answer, the pipeline goes red before users notice.

Part C: Check yourself

- Why should the same image be deployed to dev and prod?
- What is the difference between /health and /ready?
- When would you choose Container Apps over App Service?
- Where are deployment approvals configured in Azure DevOps?
- How would you roll back a bad release?

Part D: Practice ideas for tonight

- Run the full stack (API, web and database) with docker compose.
- Add a smoke test stage to your pipeline YAML.
- Write a one-page runbook: how to deploy, verify, roll back and clean up.

Coming up on Day 20: a full day of FDE professional skills: handling ambiguity and customer-embedded problem solving.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today